



FIRST SEMESTER 2026-2027

Course Handout

Date: 3 August 2026

In addition to Part I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : MAC F311

Course Title : ALGEBRA I

Instructor-in-charge : RAHUL KUMAR (kumar.rahul@pilani.bits-pilani.ac.in, 6067G)

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1. **Course description:** Groups, subgroups, a counting principle, normal subgroups, and quotient groups, Cayley's theorem, automorphisms, permutation groups, Sylow's theorems, Rings, ring of real quaternions, ideals and quotient rings, homomorphisms, Euclidean rings, polynomial rings, and polynomials over the rational field.

2. **Scope and Objective of the Course:**

The objective of this course is to provide an introduction to abstract algebraic structures, including groups and rings. The abstract setting allows for the study of disparate mathematical objects together. We can derive properties and structures that apply to concrete examples. We will see examples from number theory, linear algebra and geometry.

3. **Course learning outcomes:**

CLO1. Understand and apply the fundamental concepts of groups to solve algebraic problems

CLO2. Analyze the structure of rings, integral domains and fields using algebraic theorems

CLO3. Construct rigorous mathematical proofs to solve computational and theoretical problems

4. **Textbook:** I. N. Herstein, *Topics in Algebra*, 2nd ed., John Wiley (1999)

5. **Reference Books:**

1. Michael Artin, *Algebra*, 1st edition, Prentice Hall of India (1991)
2. David S. Dummit & Richard M. Foote, *Abstract Algebra*, 2nd edition, John Wiley (1999)
3. John B. Fraleigh, *A First Course in Abstract Algebra*, 7th edition, Pearson (2003)
4. Joseph Gallian, *Contemporary Abstract Algebra*, 8th edition, Brooks/Cole, Cengage learning (2012)





6. Lecture Plan

Module Number	Lecture session	Sections	Learning Outcome
1. Introduction to Groups and their properties	L1.1-1.3 Definition and Examples of Groups	2.1, 2.2	Understanding the concept of a group
	L1.4. Preliminary Lemmas	2.3	
2. Introduction to subgroups and a special class of subgroups	L2.1-2.2 Subgroups	2.4	Concept of subgroup has more implications than a subset.
	L2.3 A Counting Principle	2.5	
	L2.4 -2.5 Normal subgroups	2.6	Generalization of modular arithmetic to arbitrary groups
	L 2.6 Quotient groups		
3. Structure preserving maps between two groups	L3.1 -3.2 Homomorphisms	2.7	Which re-labeling of group elements is allowed?
	L3.3 Automorphisms	2.8	
4. Abstract groups are not that abstract after all	L4.1 Cayley's theorem	2.9	To learn how to extend group structure to finite permutation groups and applications of Cayley's theorem
5. An important group in algebra	L5.1-5.3 Permutation Groups	2.10	Knowledge about the permutation group and its subgroups
6. Class equation of a group, applications and Sylow's theorems	L6.1-6.2 Another counting principle	2.11	To learn applications of class equation. Apply Sylow's theorems to analyze the structure of groups and to rule out existence of simple groups of certain orders
	L6.3-6.4 Sylow's theorems	2.12	
7. Introduction to Rings	L7.1 Definition & Examples of Rings	3.1	Basic concepts of a





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	L7.2 Ring of real Quaternions	3.2	ring
8. Structure preserving maps between two rings	L8.1-8.2 Homomorphisms	3.3	Which maps between rings relate their structures?
9. Construction of Quotient rings and their properties	L9.1-9.3 Ideals and Quotient rings	3.4, 3.5	To learn modular arithmetic in rings
10. Important rings	L10.1-10.5 Euclidean rings and Principal Ideal Rings	3.7, 3.8	Algebraic properties of these rings, examples, irreducibility of polynomials over a field and construction of a field
	L10.6-10.7 Polynomial rings	3.9	
11. Rings in which factorization is a reliable process	L11.1- 11.4 Factorization of polynomials, Unique Factorization Domains	3.10, 3.11	Irreducibility of polynomials with rational coefficients, algebraic properties of unique factorization domain

7. Evaluation Scheme:

	Evaluation Components	Weightage (%)	Time (Minutes)	Date of Evaluation	Nature of Component
1	Mid-semester Exam	35%	90	As per Timetable	Closed Book
2	Continuous Evaluation (Quizzes)	20% (Best 2 out of 3 announced quizzes)	30 minutes (each quiz)	TBA	Closed Book
3	Comprehensive Exam	45%	180	As per Timetable	Closed/Open Book



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8. **Criterion for NC:** Any student scoring less than 10% marks (20 marks out of 200) will be given NC Grade. Further, if a student is absent in any evaluative component (that is, either in mid-semester exam or in comprehensive exam or in all quizzes), then that student will be given NC grade.

9. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.

10. **Chamber Consultation Hour:** To be announced in class.

11. **Notices:** All notices regarding MAC F311 will be displayed on NALANDA and/or information will be passed on in lectures/tutorials.

12. **Makeup Policy:** Make-up for the mid-semester/comprehensive examination will be given to genuine cases with prior permission only. Quizzes will not have any make-ups. No requests for make-make-up will be entertained.

(Rahul Kumar)

Instructor In-charge

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